

Zihan Nie

[Homepage](#)

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Gender: Female **Date of Birth:** 07/10/2002

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Education

Shandong University

Master of Engineering in Mechanical Engineering

Expected June 2027

Cumulative GPA (current): 88.2/100

Bachelor of Engineering in Mechanical Design, Manufacturing and Automation

June 2024

Cumulative GPA: 3.89/4.0

Publications and Patents

Published

- Z Nie, M Xu, Y Cui, H Wei, W Yi, S Niu, Y Wan, X Wei, W Song. "Few-shot Medical Anomaly Detection through Centroid Consultation Back and Test-Time Self-Calibration". *Pattern Recognition (PR)*, 2026.
- M Xu, Z Nie, B Fu, Z Chen, Z Li, H Wei, Y Wan, W Song. "Beyond Feature Mapping: Dual-Heterogeneous Knowledge Distillation with Mamba for Industrial Anomaly Detection". *Expert Systems with Applications (ESWA)*, 2026.
- M Xu[†], L Qu[†], Z Nie, F Li, Z Chen, Q Liao, Y Wan, S Niu, R Cong, X Li, W Song. "Towards reliable medical unsupervised anomaly detection: A benchmark and dataset for PET/CT with cross-modality knowledge distillation network". *Information Fusion (IF)*, 2026.
- M Xu, H Wei, Z Nie, X Zhou, B Fu, H Yan, Y Wan, W Song, "Hybrid Dual-Heterogeneous Knowledge Distillation Network for Anomaly Detection in Retinal OCT Images". *IEEE Journal of Biomedical and Health Informatics (JBHI)*, 2025.

Manuscript Under Review

- Z Nie, M Xu, Y Cui, H Wei, S Niu, Y Wan, X Wei, W Song. "CFR-Net: Collaborative Feature Refinement Network for Medical Image Anomaly Detection". *British Machine Vision Conference (BMVC)*.

Chinese Invention Patents

- W Song (Advisor), Z Nie, Y Wan, X Liang, J Wang, Z Li, K Yang, Y Zhao, Y Lv. "A Wear Error Compensation System for Wheel Diameter Gauge Positioning Blocks". China Patent ZL202410079723.9, granted November 2024.

Research Experience

Intelligent Robotics and Imaging Systems Engineering Laboratory

Shandong University

July 2023 - Present

- Advised by: Professors [Weiyue Song](#) and [Yi Wan](#)
- Research:
 - Developed multiple **few-shot and unsupervised anomaly detection** frameworks with applications to medical imaging, focusing on **VLM representation** and score calibration, cross-space consistency-guided **teacher-student distillation**, and **multimodal imaging fusion**.
 - (1) Designed a **CLIP-based few-shot** anomaly detection framework with **Centroid Consultation Back** and

Test-Time Self-Calibration to improve representation robustness and decision reliability.

(2) Proposed CFR-Net, an **unsupervised** anomaly detection framework that **advances teacher-student distillation** through input-side collaborative feature refinement, output-side cross-space consistency learning, shared-parameter MPFRM, and variance-sensitive optimization.

(3) Proposed a **multimodal** anomaly detection framework based on OCT/OCTA, leveraging hierarchical vascular information (superficial and deep flow) with a novel cross-modal fusion strategy to enhance detection performance.

- Contributed to the development of high-resolution **OCT imaging systems**, including visible and near-infrared OCT and endoscopic OCT. Participated in system simulation, prototype development, and experimental validation of the first 550 nm visible-light endoscopic OCT system (5 μm lateral / 1.33 μm axial resolution) Supported functional imaging capabilities such as quantitative measurement of retinal oxygen saturation (sO₂) and capillary-level blood flow velocity.

Cardiovascular Imaging Intelligent Analysis Research Group

Shandong University & Beihang University (BUAA)

February 2023 - September 2024

- Advised by: Professors [Jie Tian](#), [Enqing Dong](#), [Zhenchao Tang](#), and [Peng Xue](#)
- Research:
 - Spearheaded artery tree **segmentation** from **Coronary CT Angiography (CTA) volumes** for quantitative analysis of 3D cardiovascular imaging, designing an end-to-end deep learning pipeline with customized segmentation architecture, enhanced multi-scale feature extraction, and topology-preserving post-processing.
 - Achieved a **Dice score of 0.837**, demonstrating strong effectiveness for quantitative cardiovascular image analysis.

Professional Skills

- **Programming Languages:** Python, C
- **Deep Learning Frameworks:** PyTorch
- **Models & Techniques:** Vision-language models, Transformer-based models
- **Image Analysis & Annotation Tools:** DarkLabel, ITK-SNAP, 3D Slicer
- **Simulation & Design Software:** Abaqus, ANSYS, Zemax, SolidWorks

Research Interests

- Vision-language models for visual understanding and reasoning
- Multimodal visual representation learning and feature fusion
- Robust and generalizable visual AI for real-world visual decision-making

Honors and Awards

- **First-class Academic Scholarship**, Shandong University, 2025, 2023
- **Graduate Freshman Scholarship**, Shandong University, 2024
- **Second-class Academic Scholarship**, Shandong University, 2022
- **Third-class Academic Scholarship**, Shandong University, 2021
- **Social Practice Scholarship**, Shandong University, 2021

Hobbies

- Music, fitness, baking, and swimming